Concept of handling GPS, collecting ground control points and referencing

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SMUSDP Training

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Global Positioning System(GPS)

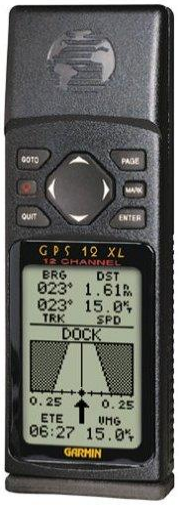
Global : Anywhere on Earth !

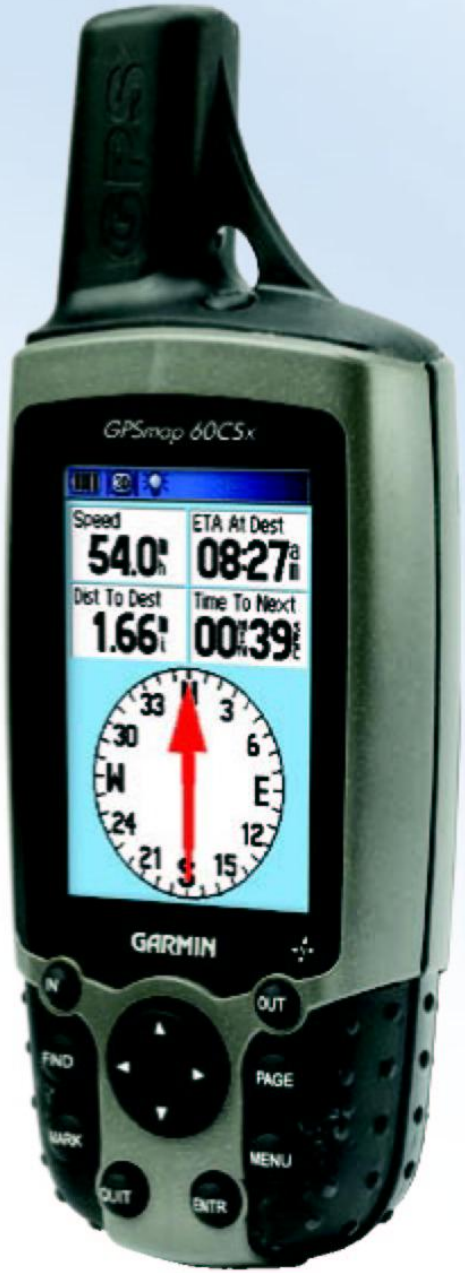
Positioning : Where are you ? How far you are moving and in what direction ? What direction should you go to get some other specific location, and how long does it take at your speed to get there ? And most importantly for GIS, where have you been ?

System : A collection of components with connections(links) among them.

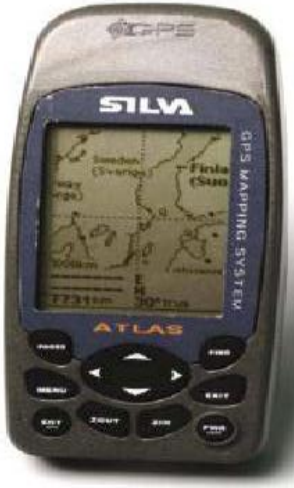
What is GPS ? Brief Introduction :

- Satellite-based Radio Navigation System designed to provide accurate positioning information.
- Originally GPS was developed as a defense system by US Department of Defense to simplify accurate navigation.
- Now, this satellite system is authorized to use widely by Civilian in worldwide



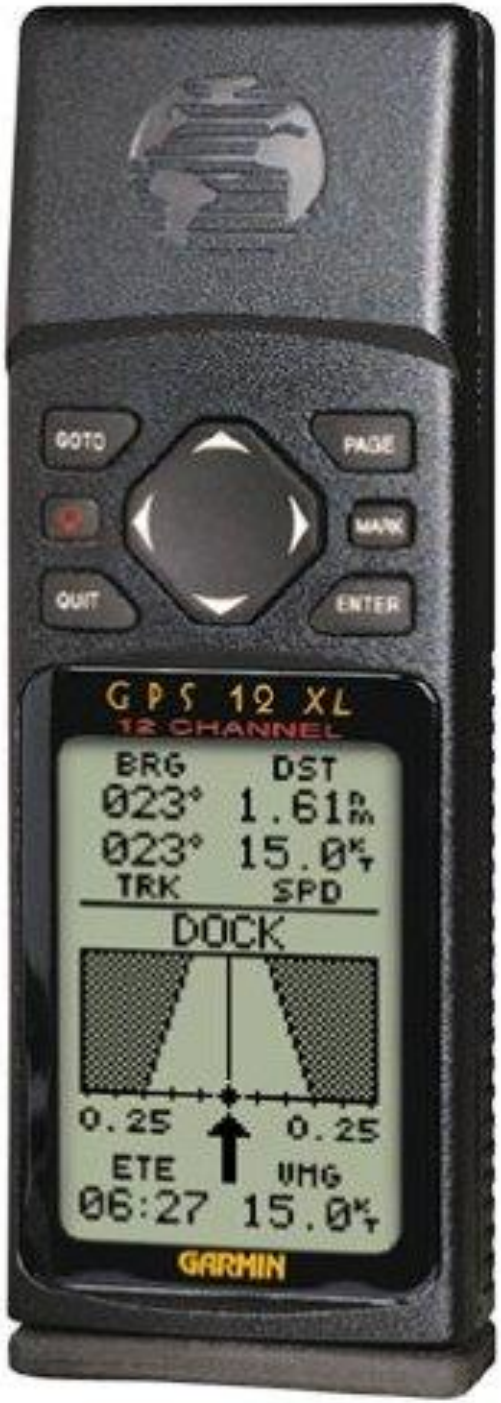

GPSmap 60CSx
Speed
54.0
ETA At Dest
08:27
Dist To Dest
1.66
Time To Next
00:39
GARMIN
IN
OUT
FIND
PAGE
MARK
MENU
QUIT
ENTR

MULTI NAVIGATOR
BACKTRACK
COMPASS
DISTANCE
2748 KM
SILVA


SILVA
GPS MAPPING SYSTEM
ATLAS

SILVA
COMPASS
KM 2748

page
goto
on/off
rocker switch
mark
screen (map page)


GOTO
PAGE
MARK
QUIT
ENTER
GPS 12 XL
12 CHANNEL
BRG DST
023° 1.61
023° 15.0
TRK SPD
DOCK
0.25 0.25
ETE VMG
06:27 15.0
GARMIN

GPS
12XL
GARMIN

Garmin GPSMAP 478

Compatible with ExpertGPS

How GPS Works?

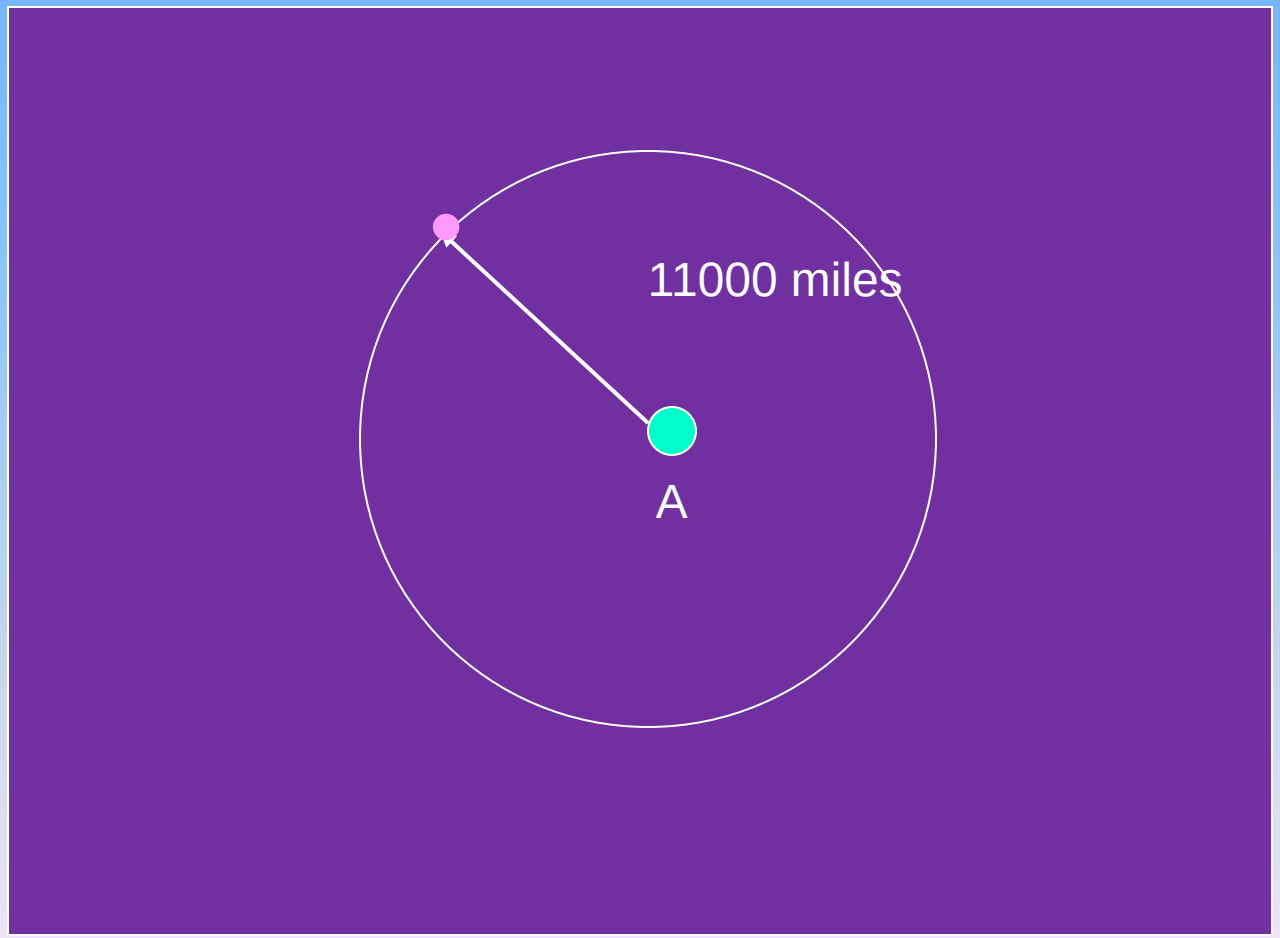
- GPS system works by timing how long it takes a radio signal to reach us from a satellite and then calculating the distance from that time.
- To measure travel time, GPS needs very accurate clocks.
- Radio waves travel at the speed of light: 186,000 miles per second.

- **If we can figure out exactly when the GPS satellite started sending its radio message and when we received it, we will know how long it took to reach us.**
- **We just multiply that time in seconds by 186,000 miles per second and that's our range to the satellite.**

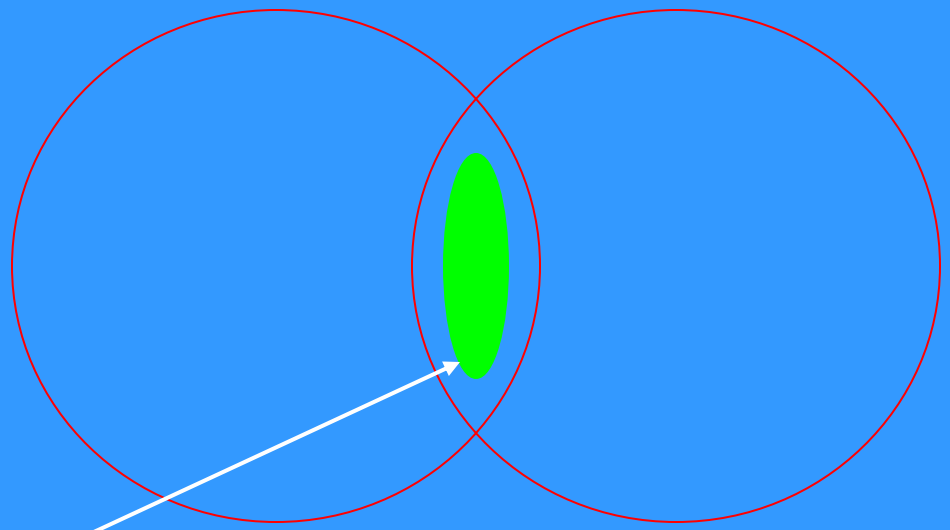
What is satellite Ranging?

- **Satellite Ranging means that we figure our position on earth by measuring our distance from a group of satellites in space.**
- **The satellite acts as precise reference points for us.**

Let us say we are at 11,000 miles from Satellite A.

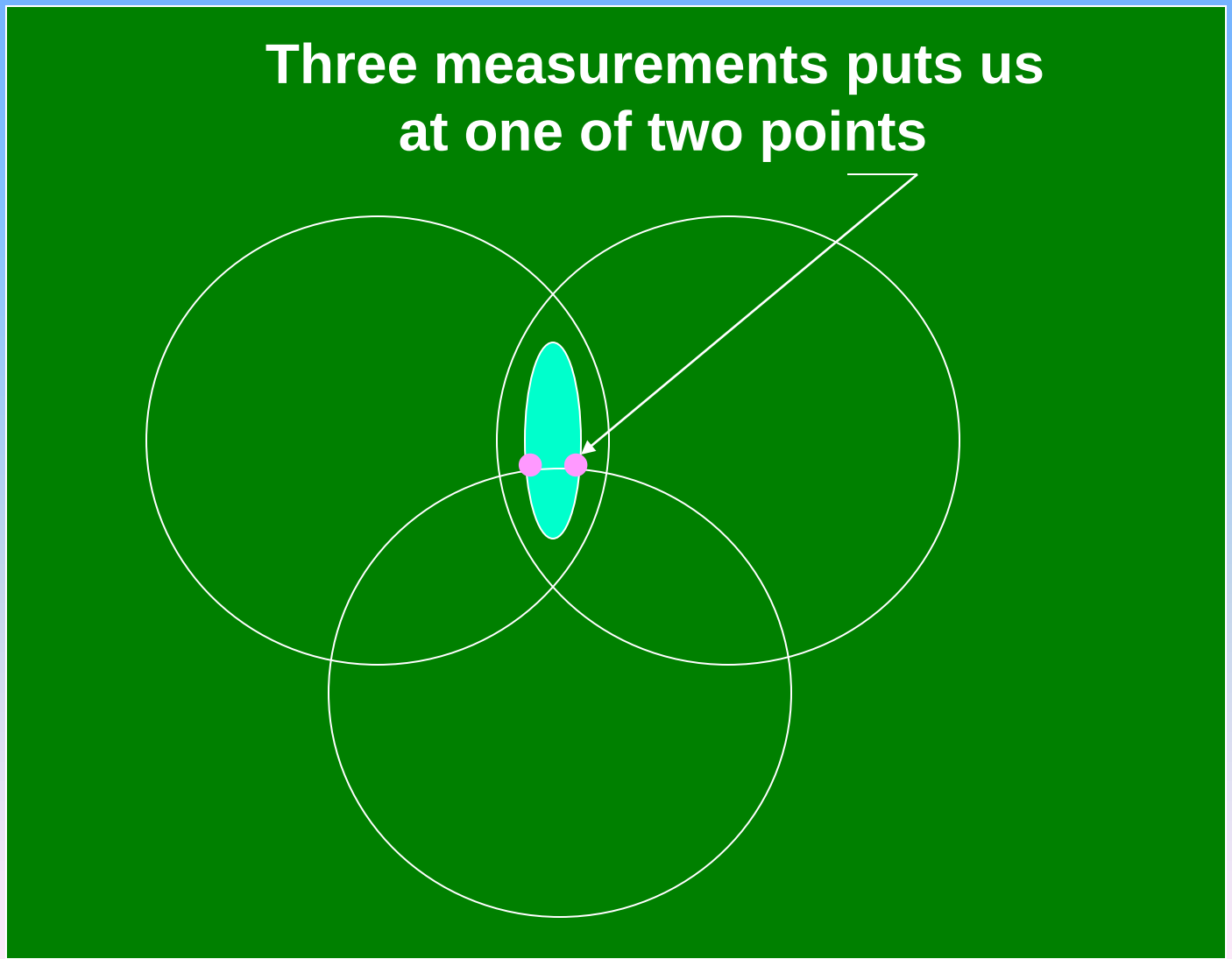


At the same time, let us say we are at 12,000 miles from Satellite B.



Two Measurements puts us
somewhere on this circle
(where two spheres intersect)

At the same time, if we are 13,000 miles from Satellite C, there are only two points in space where that can be true.



- Usually, one of the two points is a ridiculous answer.
- The computers in GPS receivers have various techniques for distinguishing the correct point from the incorrect one.
- Anyway, if we wanted to be absolutely technical, trigonometry says we really need four satellite ranges to unambiguously locate ourselves.

What is Differential GPS?

- It is the technique of boosting the accuracy of GPS.
- With it, GPS can achieve measurement accuracies of better than a meter.
- If we put a GPS receiver on the ground in a known location (Benchmark), we can use it to figure out exactly what errors the satellite data contains.
- It can then transmit an error correction message to any other GPS receivers that are in the local area, and they can use that error message to correct their position solutions.

Why GPS ?

- **Location** :
  To determine absolute location
- **Navigation** :
  Getting from one location to another
- **Tracking** : Monitoring the movement of people and other things
- **Mapping** :
  Creating maps of the World
- **Timing** :
  Bringing precise timing to the World

GPS (Global Positioning System)

Determines Real world coordinate

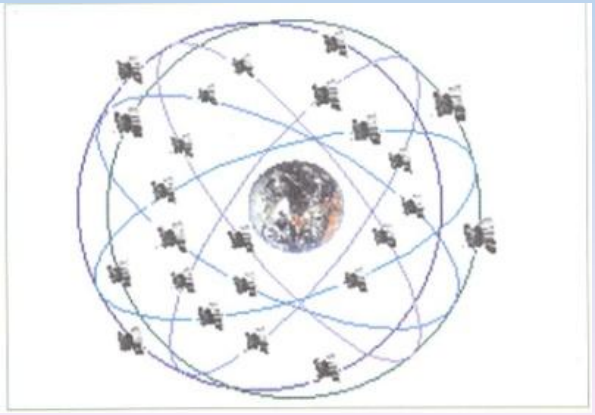
* Longitude
* Latitude
* Altitude
* Time

Three segments of GPS

- **Space segment** : consisting of satellites which broadcast signals
- **Control segment** :

  steering the whole system
- **User segment** :

  including many types of receivers



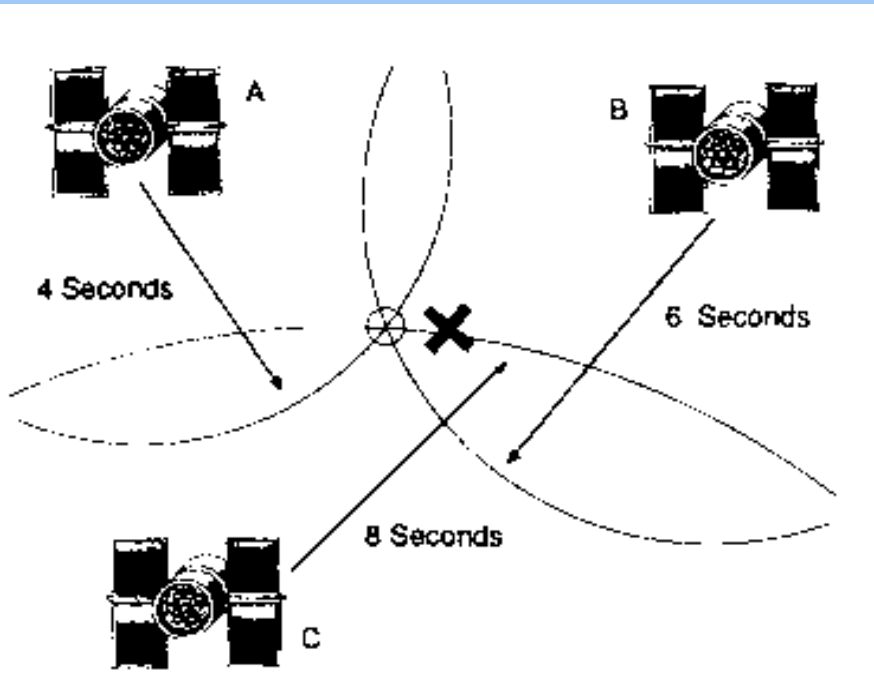
SPACE SEGMENT

- 24 Satellites
- 6 Orbital planes, 4 satellites on each orbit in equally spaced
- 55 degree inclination to equatorial plane
- 20,200 Km(10,900 N. Miles) altitude
- 24 Hour coverage 3-D Worldwide
- 12 Hour Orbital period
- Velocity of satellites is approx. 4 km/sec.
- 5-8 Visible Satellites from any point

For the GPS Position

* 4 satellites for 3 D.

* 3 satellites for 2 D.



eTrex Description

GARMIN GPS

Control Buttons

Antenna
Internal

Rocker Button
Up goes into alphabet
Down goes into numbers

Go To Button
Press once, it brings up list of waypoints.
Select (highlight) the way point and press ENTER

Page Button
Cycles through the 5 pages

Mark Button
Press once, it records location, you enter name

On/Off Button
Hold down for 3 seconds to turn off

Enter Button
Enters a mode, or 'selects' the highlighted option

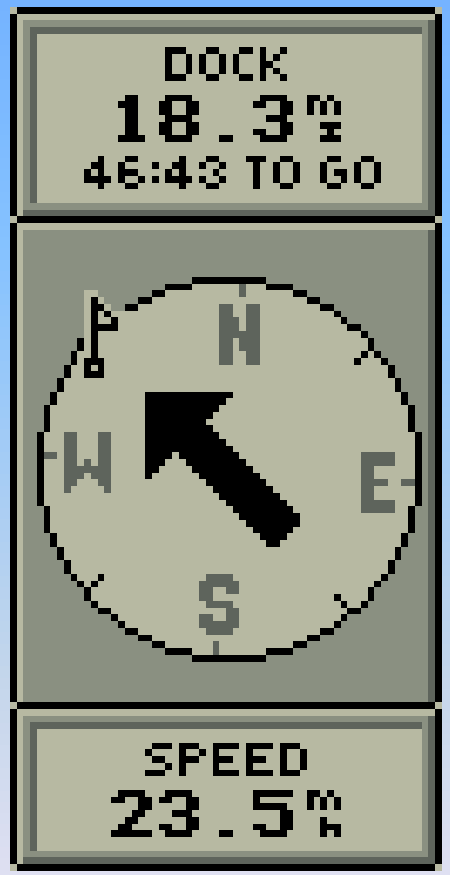
Quit Button
Cycles backward through pages, or moves backward out of a mode.

Display
Map page shown

MAIN FUNCTIONAL PAGES

MAIN MENU

MAP PAGE



POINTER PAGE

MARK POINTER PAGE

Navigation Features

Waypoints/Icons	500 with name and graphic symbol
Track log	10,000 points, 10 saved tracks
Routes	20 reversible route with up to 500 waypoints
Trip computer	Current speed, average speed, time of sunrise/sunset, resettable maximum speed, trip timer, and trip distance
Map Datums	More than 100
Position format	Lat/Lon, UTM/UPS, Maidenhead, MGRS and other grids

Parameters Setting of the GPS

* Geographic (lat/long)
  - DD/DM/DMS

  Projected/Grid (UTM)
  - Easting/Northing

* Datum - Indian/Bangladesh

* North reference -True

* Reference altitude - 0000

* Calibrate Altitude - Known point altitude

(continue ...)

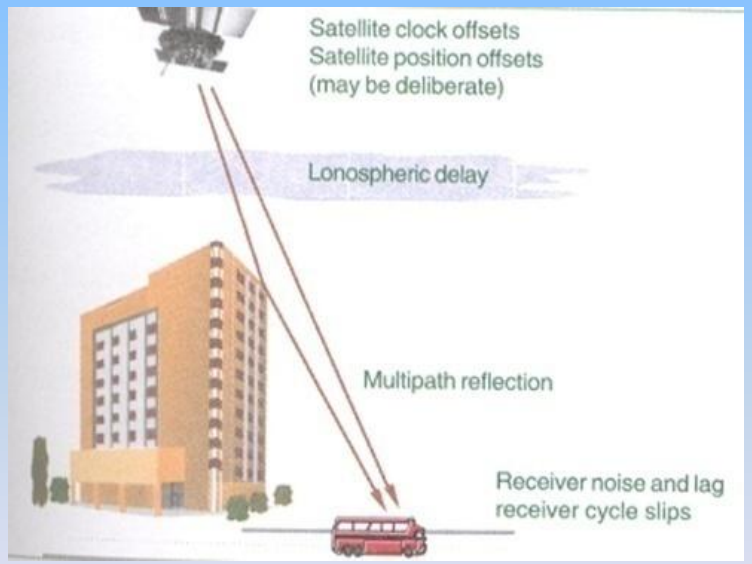
* Units
  - distance (KM)
  - speed (KM/hrs)
  - temperature (centigrade)
  - pressure (MMHG)
  - altitude (metres)

Types of data capture

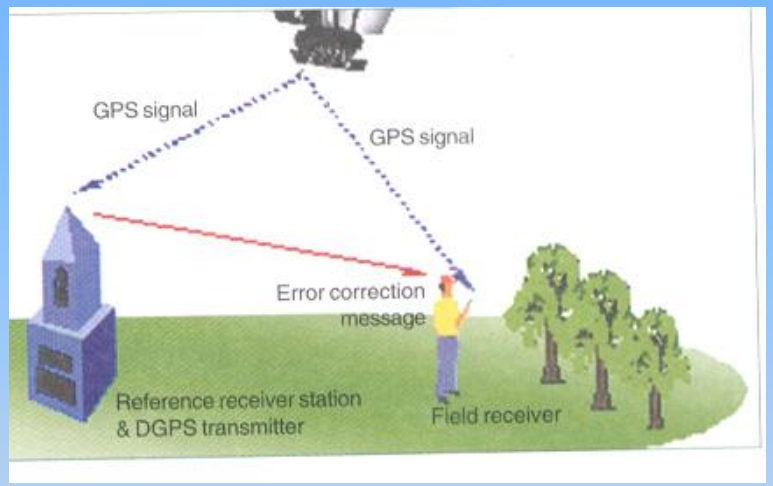
* Way point (for point)
* Track log (for line)
* Route (to find out new location)

GPS Errors

- Satellite errors
- The atmosphere
- Multipath
- Receiver
- Selective availability



Accuracy	**Position: 15 meters (49 feet) RMS***



GPS ACCURACY

* Using DGPS
* Updating parameters
* Using proper time
* Using high accuracy GPS
* Surveying accuracy and so on

Types of GPS

* Survey Grade (Leica, ….)
  - mm accuracy
  * Others (Different;

  GARMIN/ etrex /vista/ 12XL, 60CSx, etc
  Silva/Atlas,
  Trimble: Geo-Explorer II, Geo XT, Geo XH, Pathfinder …)
  Motorola: Sony: etc…..
  - 2 – 5 m. accuracy (*15 m. for etrex*)

Data download/integrated with GIS

* Download data
* Display with map
* Generate the theme
* Change in different projection

GPS Application
(Why & where we can use GPS)

- **Surveying and mapping**
- **Emergency services**
- **Military services**
- **Natural resources**
- **Transportation**
- **Urban planning and management**
- **Watershed management**
- **Mining and exploitation**

and so on…

THANK YOU for YOUR KIND ATTENTION !

&……

Lets go to field for data capture/collection

Method of Data Transferring from GPS to Arc GIS/Arc View through Map source:

- Connect GPS with PC/Laptop.
- Start Map source
- Press Transfer in Menu of Map source
- Choose Receive from Device Option
- Data (Waypoints & Tracks together) will be downloaded to PC/Laptop
- Save the data in Garmin Data Base (.gdb) format in giving file name.

- Select the required Waypoints, copy it and paste it in a new file.
- Save it in a Ta Delimitated Text (TAB) format.
- Open Arc GIS software, select Tool from the Menu, click on Add XY data (follow the usage given)
- or Open Arc View software.
- Go to File, Extension and check AV Garmin Extension and also Check Default Option.
- Click AV Garmin Button (at the farthest right of menu bar), a Sub Menu will appear.

- In GIS input, click Map source w/elevation option, in Feature Type, Click Waypoint, Click Create.
- Select Map source Waypoint File/ Text file created earlier.
- Give the new name for output shape file.
- It will appear in the view.
- Follow step 7 for Track instead of waypoint.
- In step 12 under feature type, click Track/Line, in feature option select Meter.

Thank you again !